

Multiple Choice

1. **Answer:** A

Options: A) Mimetic coloration; B) Aposomatic coloration; C) Deceptive markings; D) Batesian mimicry; E) Flash coloration

Note: Correct option: A - Mimetic coloration

2. **Answer:** A

Options: A) Tonus refers to the electrical activity in the brain during relaxation.; B) Tonus refers to the maximum force a muscle can exert in a single effort.; C) Tonus refers to the flexibility and range of motion in joints and muscles.; D) Tonus refers to the state of complete relaxation in skeletal muscles.; E) Tonus refers to the involuntary contraction of smooth muscles in the digestive tract.

Note: Correct option: A - Tonus refers to the electrical activity in the brain during relaxation.

3. **Answer:** A

Options: A) Courtship is primarily for entertainment purposes among terns; B) Courtship is not necessary for reproduction; C) Courtship is a way to show dominance; D) Courtship is a random behavior with no impact on reproduction; E) Courtship is necessary for physical transformation required for reproduction

Note: Correct option: A - Courtship is primarily for entertainment purposes among terns

4. **Answer:** C

Options: A) conjugation results in no genetic diversity; B) conjugation requires more than two organisms; C) all individuals are identical genetically, and conjugation could not occur; D) repeated fission in paramecia results in sterile offspring that cannot conjugate; E) the group of paramecia lacks the cellular structures necessary for conjugation

Note: Correct option: C - all individuals are identical genetically, and conjugation could not occur

5. **Answer:** A

Options: A) peroxisome; B) cytoplasm; C) nucleus; D) intermembrane space; E) outer mitochondrial membrane

Note: Correct option: A - peroxisome

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Golgi apparatus and endoplasmic reticulum'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '4'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Reduction of NADP+ to NADPH'.

Long Form Response

11. **Answer:** Robin, Lizard. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Bacteria are used in the dairy industry for the production of fermented milk products, butter, and different types of cheeses. They also help certain milk-giving animals acquire nutrients by breaking down cellulose from ingested plant material.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key